

Chemistry: MCQ Practice Paper

Q1. (Atomic Structure) Which quantum number determines the shape of an orbital? **a) n, b) l, c) ml, d) ms**

Q2. (Bonding) Which geometry does an sp^3 hybridized central atom typically exhibit? **a) Linear, b) Trigonal planar, c) Tetrahedral, d) Octahedral**

Q3. (Thermodynamics) For a spontaneous process at constant T and P, the change in Gibbs free energy (ΔG) is: **a) > 0 , b) < 0 , c) $= 0$, d) undefined** Q4. (Kinetics) According to the Arrhenius equation, a plot of $\ln(k)$ versus $1/T$ is a straight line with a slope of: **a) $-E_a/R$, b) E_a/R , c) A, d) $-E_a$**

Q5. (Coordination) In $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$, what is the coordination number of Cobalt? **a) 3, b) 6, c) 9, d) 12**

Q6. (Organic) Which type of isomerism is shown by 1-butene and 2-butene? **a) Chain, b) Functional, c) Position, d) Optical**